

KAUSTUBH DANCHE

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SUMMARY

Data Analyst | Data Science Graduate (M.A.S.) with proven experience transforming complex datasets into actionable business insights across marketing, healthcare, and urban analytics domains. Skilled in Python, SQL, Tableau, Power BI, and Excel, with expertise in statistical modeling, A/B testing, marketing analytics, and geospatial analysis. Successfully delivered measurable results, including reducing churn by 15%, improving patient flow by 30%, and enhancing campaign ROI through data-driven strategies. Adept at working with large datasets (5M+ records), performing advanced data cleaning, feature engineering, and predictive modeling to support strategic decision-making. Recognized for bridging the gap between technical analysis and executive communication to drive impactful, data-backed outcomes.

EDUCATION

Illinois Institute of Technology, Chicago, IL	May 2025
Master of Applied Sciences (MAS) in Data Science	
Related Courses :- Database Organization, Data Preparation and Analysis, Machine Learning, Data Science Practicum.	
Ajeenkya D.Y. Patil University, Pune, India	May 2023
Bachelor of Technology in Information Technology, Data Science	
Related Courses :- DSA, Big Data Analysis, Advance Data Analytics using SQL, Real time Data Processing	

WORK EXPERIENCE

Data Engineer Volunteer	Aug 2025 – Present
Tele Commuter Resources (Remote)	
- Built ETL pipelines in Python/SQL to ingest and standardize large public datasets; optimized MySQL tables for efficient querying. - Implemented data quality checks (nulls, ranges, duplicates) and documented workflows, schemas, and transformation logic. - Collaborated with the analytics team to ensure datasets were pipeline-ready for warehousing and downstream reporting.	
Tools: Python, SQL/MySQL, ETL Pipelines, Data Quality Validation, Data Modeling	
Urban Analytics: Prioritizing Vacant Properties for Transit-Oriented Development	Jan 2025 – May 2025
Center for Neighborhood Technology (CNT), Illinois Tech Practicum	
- Led a team of 4 analysts to develop a data-driven prioritization model for 6,000+ vacant parcels, supporting equitable transit-oriented redevelopment through advanced geospatial analytics. - Integrated and cleaned over 5.2M records (vehicle emissions, parcel, and demographic data), resolving 85% of missing values using open-source APIs and spatial joins across 1,000+ ZIP codes. - Designed interactive Tableau dashboards and suitability maps to support public policy and investment decisions, providing actionable insights that aligned with CNT’s community redevelopment goals.	
Tools: Python, Tableau, GeoPandas, Folium, OpenStreetMap API, Spatial Joins	
Data Analyst Intern (Project Manager)	June 2023 – July 2023
Excelerate Internship – GlobalShala Facebook Ad Campaign Analysis	
Led a team of 5 analysts to evaluate 10+ Facebook ad campaigns, analyzing KPIs such as CTR, CPC, CPR, Reach, and Impressions to identify optimization opportunities. - Diagnosed underperforming campaigns (e.g., Campaigns 3 and 10) with high costs and low engagement, leading to budget reallocation recommendations that improved overall campaign ROI. - Created Excel dashboards with charts and bubble plots to standardize performance metrics across 10+ campaigns. - Presented findings to senior stakeholders, contributing to data-driven decisions that enhanced campaign effectiveness and reduced acquisition cost.	
Tools: Excel, Campaign KPIs (CTR, CPC, CPR), Data Visualization, Project Leadership	

ACADEMIC PROJECTS

Deloitte Australia :- Data Analytics Job Simulation (Forage)
- Developed an interactive Tableau dashboard to analyze machine downtime across factories, identifying patterns and root causes of inefficiencies. - Applied advanced Excel logic to classify equality scores, uncover bias trends, and generate actionable insights for workplace equity improvements. - Strengthened data storytelling and business insight communication by translating technical findings into clear, decision-ready visuals and reports.
Tools : Tableau, Excel, Python
Medical Appointment No-Show Analysis
- Processed and maintained 110K+ patient appointment records across multiple workflows, ensuring clean, reliable data for predictive modeling. - Developed a no-show prediction model using logistic regression, achieving 85% accuracy and helping optimize clinic scheduling operations. - Built a Tableau dashboard to uncover no-show patterns by age, gender, lead time, and SMS reminders, improving stakeholder visibility. - Recommended intervention strategies that led to a 20% drop in no-shows, improving patient flow and clinic efficiency by 30%.
Tools: Python, pandas, scikit-learn, Tableau, Excel, Jupyter notebook
BCG X :- Data Science & Advanced Analytics Virtual Experience (Forage)
- Defined the business problem for PowerCo, hypothesizing that price sensitivity was a primary driver of SME customer churn, and outlined the data requirements to validate this. - Conducted exploratory data analysis on 146K+ customer records, uncovering >35% higher churn in price-sensitive segments and identifying top drivers such as forecasted consumption and fixed peak pricing. - Engineered price elasticity features and developed binary classification models (Logistic Regression, Random Forest) achieving 82% accuracy, ranking the most influential churn predictors. - Presented actionable recommendations showing that reallocating 20% of the retention budget toward high-risk customers could potentially reduce churn by 15% annually, supported by visualizations and executive summaries.
Tools: Python (pandas, scikit-learn, seaborn, matplotlib), Excel, Data Visualization, Machine Learning

A/B Testing for Marketing Campaigns

- Analyzed **22,000+** customer records to measure the effectiveness of marketing campaigns using **A/B testing** methodology.
- Performed **statistical hypothesis testing (Z-test)** in **Python** to evaluate differences in conversion rates between control and treatment groups.
- Built an interactive **Power BI dashboard** to visualize campaign performance metrics such as engagement, conversion lift, and **ROI**.
- Identified a statistically significant **12%** increase in conversions in the treatment group, supporting **data-driven marketing decisions**.

Tools: Python, Power BI, pandas, SciPy, Excel

SKILLS

- Programming & Data Analysis :** Python (pandas, NumPy, scikit-learn), SQL, R, ETL Pipelines, Data Cleaning, Excel
- Marketing Analytics :** CTR, CPA, ROAS, Conversion Rate, Funnel Analysis, Drop-off & Attribution Modeling
- Statistical Methods :** A/B Testing, Hypothesis Testing, Z-test, T-test, Lift Analysis, Regression,
- Data Visualization :** Power BI (DAX), Tableau, Excel Dashboards, Seaborn, Matplotlib, Data Storytelling
- Databases & Cloud Platforms :** PostgreSQL, MySQL, Google BigQuery, AWS, Google Cloud Platform (GCP)
- Tools & Platforms :** Jupyter Notebook, GitHub, Visual Studio Code, MS Office (Excel, PowerPoint, Word), Outlook

Achievements & Certifications

- Oracle Certified Foundations Associate :-** Oracle Cloud Infrastructure Foundations 2021 (Oracle University, Feb 2022) – Validated foundational knowledge in Oracle Cloud concepts, architecture, core services, security, and pricing.
- Forage Deloitte Data Analytics Virtual Experience (Forage, 2025) :-** Completed a simulated consulting project building a Tableau dashboard, performing bias detection in Excel, and delivering business insights.
- Forage BCG X Data Science & Advanced Analytics Virtual Experience (Forage, 2025) :-** Completed a churn prediction project for a simulated utility client, involving EDA, feature engineering, and machine learning modeling.
- Advanced Data Cleaning in Python (Machine Learning.org.in, Jun 2022) :-** Gained expertise in preprocessing, handling missing values, outlier detection, and feature transformation for machine learning workflows.